

Mohd Arif Ansari

+91-9779383344

arifpu786@gmail.com

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Career Highlights

AI Engineer with 3.5+ years of experience building **scalable AI systems and data platforms**, combining strong expertise in **Generative AI (RAG, Agentic AI, LLM fine-tuning)** with **production-grade data engineering**. Proven track record of developing **end-to-end ML pipelines**, real-time data systems, and **LLM-powered applications** with measurable improvements in accuracy, latency, and cost. Skilled in bridging **data infrastructure with AI models**, enabling reliable and efficient deployment in real-world environments.

AI/ML & Data Engineering Stack

AI/ML & Generative AI: PyTorch, Scikit-learn, Transformers (BERT, LayoutLM), LLMs (OpenAI, Gemini, LLaMA), Prompt Engineering RAG, Agentic AI (Multi-Agent Systems), Fine-tuning (LoRA, PEFT), Embeddings, Vector Databases, MLflow

Data Engineering & Cloud: Apache Spark, PySpark, Structured Streaming, Delta Lake, Kafka, ETL/ELT Pipelines, Data Modeling, Data Validation, Azure Databricks, ADLS Gen2, Azure Data Factory, Synapse Analytics, PostgreSQL, Snowflake, SQL Server, NoSQL

Programming & Tools: Python (Pandas, NumPy), SQL, FastAPI, Docker, CI/CD (GitHub Actions, Azure DevOps), Power BI

Professional Experience

Data Engineer | Ingenious e-Brain ,06 Jan 2025 – 23 May 2025

- Built distributed PySpark pipelines on Azure Databricks processing **10M+ records/week**; improved job runtime from **10 min to 6 min (40%)** through partitioning, caching, and optimized joins.
- Developed real-time Kafka + Spark streaming pipelines handling **5K events/min**, reducing end-to-end latency from **3–5 sec to < 500 ms (85%)** by tuning micro-batch intervals, eliminating data skew, and using broadcast joins.
- Implemented Delta Lake architecture (ACID, schema evolution, time travel), reducing data inconsistency issues and enabling reliable incremental processing.
- Designed low-latency data pipelines for analytics, improving dashboard freshness from **minutes to sub-second updates**.
- Optimized Spark workloads using partition pruning, predicate pushdown, and memory tuning, increasing throughput by **35–40%** compared to baseline jobs.
- Automated ETL pipelines with validation checks, reducing data quality issues (nulls, schema mismatch) observed in earlier pipelines.
- Integrated pipelines with Snowflake and ADLS; enabled CI/CD-based deployment for reproducible production workflows.

Data Analyst | Provana India , 07 Jan 2021 – 30 Dec 2024

- Built scalable ETL pipelines using PySpark and Snowflake, reducing data processing time from **8 hours to 4.5 hours (40%)** through query optimization and parallelization.
- Designed and implemented data warehouse pipelines, improving query latency for analytics workloads by **30–40%**.
- Developed multi-source data integration frameworks, replacing manual data stitching with automated pipelines.
- Implemented data validation and governance checks, significantly reducing inconsistencies observed in legacy reporting systems.
- Optimized complex SQL queries (joins, indexing strategies), improving execution efficiency across reporting pipelines.
- Delivered ML-ready datasets with consistent schema and feature engineering support for downstream data science use cases.

Key Projects

Personal AI Chatbot (RAG + Self-RAG System)

- Built a RAG-based chatbot pipeline: **ingestion** → **chunking** → **embeddings** → **retrieval** → **generation** → **validation loop** using FAISS and LLM APIs.
- Baseline:**
 - Standard RAG (top-k=3, no validation)
 - Hallucination rate: **~40%** (answers not grounded in retrieved context)
 - Retrieval success rate: **~55%**
- Improvements:**
 - Implemented **Self-RAG validation loop** (LLM-based grounding check + confidence scoring)
 - Added **re-retrieval strategies** (query reformulation + dynamic top-k)
 - Enforced grounded answer generation
- Results:**
 - Reduced hallucination rate from **40% → 26% (30–35%)**
 - Improved retrieval success rate from **55% → 69% (25%)** via top-k tuning (3 → 5)
- Enabled multi-turn conversations with memory handling and context tracking.
- Built modular FastAPI services with Docker-ready deployment.

Tech: LangChain, FAISS, Python, FastAPI, LLM APIs

LLM Fine-Tuning & Optimization Pipeline

- Fine-tuned LLM using **LoRA (PEFT)** for domain-specific QA tasks.
- **Baseline:**
 - Zero-shot / prompt-engineered LLM
 - Accuracy: **62%**, F1: **0.68**
- **Improvements:**
 - Applied LoRA (low-rank adapters) to attention layers with domain dataset
 - Fine-tuned only **1–5% parameters** instead of full model
- **Results:**
 - Accuracy improved to **78% (20–25%)**, F1 to **0.82**
- **Cost Optimization:**
 - Reduced training cost by **70–80%** (full fine-tuning → PEFT)
 - Lower GPU memory usage and faster training cycles
- **Inference Optimization:**
 - Reduced latency from **1s** → **600–700 ms** via prompt compression
 - Implemented caching (query + embedding), reducing repeated query latency to **~100 ms**
 - Improved throughput **2–3x** using async and batch processing
- **Model Routing System:**
 - Routed queries to small/medium/large models based on complexity
 - Reduced inference cost by **50–55%** compared to single-model baseline

Tech: HuggingFace, PEFT (LoRA), Python, FastAPI

Computer Vision System (Medical Image Classification)

- Built EfficientNet-based model for **benign vs malignant classification** with Grad-CAM explainability.
- **Baseline:**
 - CNN trained from scratch
 - AUC: **0.75–0.78** (limited data, overfitting)
- **Improvements:**
 - Applied **transfer learning (EfficientNet)** pretrained on ImageNet
 - Fine-tuned higher layers for domain adaptation
 - Introduced **data augmentation** (rotation, flip, crop, jitter)
 - Handled class imbalance using weighted loss
- **Results:**
 - Improved AUC to **0.88–0.90 (12–15%)** on patient-level split
- Integrated **Grad-CAM** for visual explainability, enabling clinical interpretability of predictions.
- Built full pipeline: preprocessing → training → evaluation → visualization

Tech: PyTorch, OpenCV, EfficientNet, Grad-CAM

Certifications: - Microsoft Certified: Azure AI Engineer, Data Scientist, Data Engineer, Data Analyst

Education

Master of Technology in AI & ML, Lovely Professional University.	2025 - Present
Postgraduate in Advance Data Science and ML, IIT Madras.	2023 - 2024
Bachelor of Technology in AI & ML, Lovely Professional University.	2016 - 2020